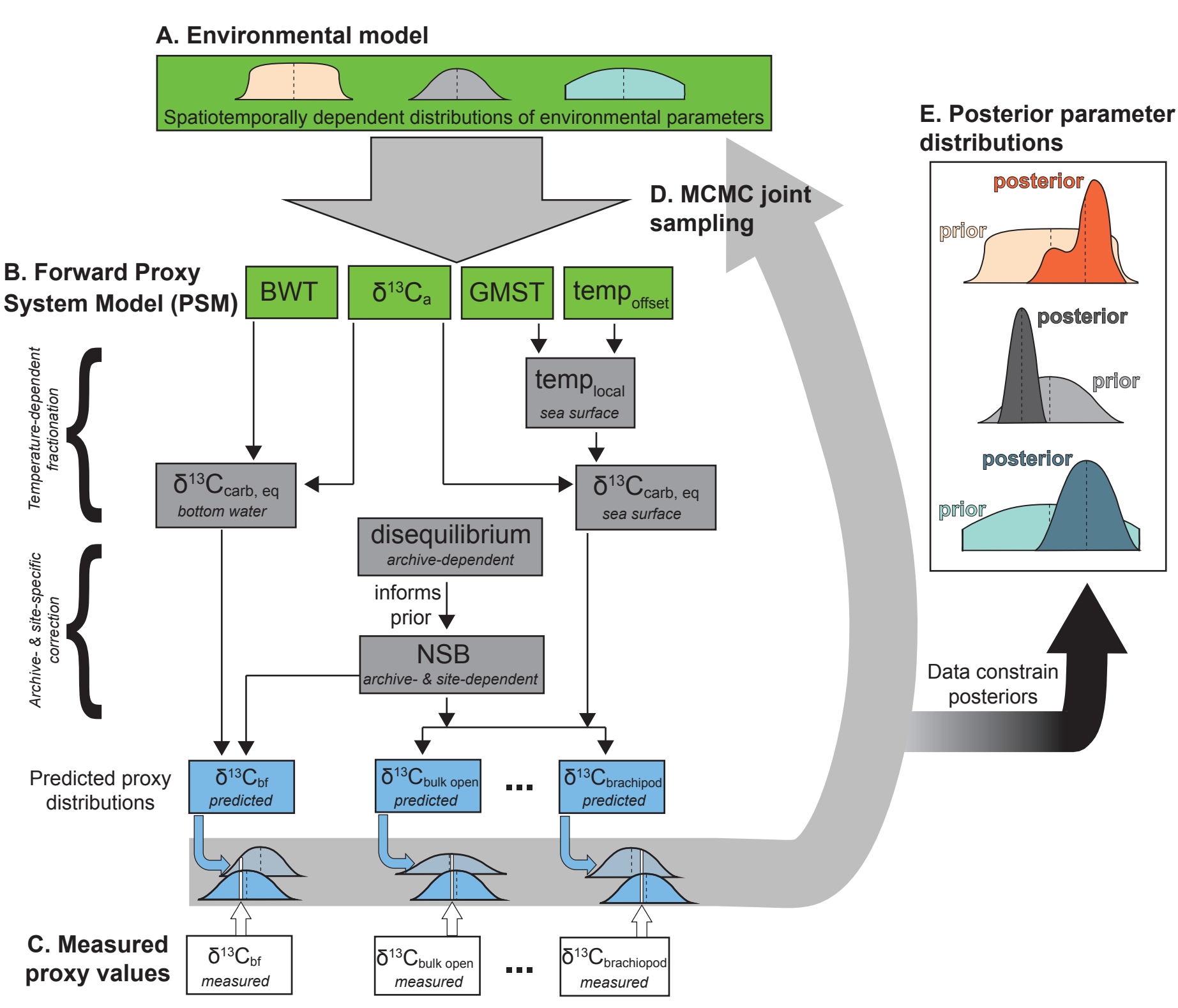

A. Environmental model
Spatiotemporally dependent distributions of environmental parameters
D. MCMC joint sampling
E. Posterior parameter distributions
posterior
prior
posterior
prior
posterior
prior
B. Forward Proxy System Model (PSM)
BWT
δ13Ca
GMST
tempoffset
templocal
sea surface
Temperature-dependent fractionation
δ13Ccarb, eq
bottom water
δ13Ccarb, eq
sea surface
disequilibrium
archive-dependent
informs prior
NSB
archive- & site-dependent
Archive- & site-specific correction
Data constrain posteriors
Predicted proxy distributions
δ13Cbf
predicted
δ13Cbulk open
predicted
...
δ13Cbrachiopod
predicted
C. Measured proxy values
δ13Cbf
measured
δ13Cbulk open
measured
...
δ13Cbrachiopod
measured